

```
#!/bin/bash
echo "Enter two no"
read a b
echo " add $((a+b)) "
echo " sub $((a-b)) "
echo " mul $((a*b)) "
echo " div $((a/b)) "
echo " mod $((a%b)) "
```

```
[sudiksha@localhost ~]$ vi arith.sh
[sudiksha@localhost ~]$ bash arith.sh
Enter two no
5 10
add 15
sub -5
mul 50
div 0
mod 5
[sudiksha@localhost ~]$
```

```
#!/bin/bash
read year
if [ $((year%4)) -eq 0 ] || ([ $((year%400)) -eq 0 ] && [ $((year%100)) -ne 0 ])
then
    echo " $year is a leap year "
else
    echo " $year is not a leap year "
fi
```

```
[sudiksha@localhost ~]$ vi leap.sh
[sudiksha@localhost ~]$ bash leap.sh
2005
2005 is not a leap year
[sudiksha@localhost ~]$ bash leap.sh
2024
2024 is a leap year
[sudiksha@localhost ~]$
```

```
#!/bin/bash
echo " Enter a number "
read num
rev=0
while [ $num -gt 0 ]
do
    rem=$((num%10))
    rev=$((rev*10+rem))
    num=$((num/10))
done
echo " Reversed number: $rev "
```

```
[sudiksha@localhost ~]$ vi reverse.sh
[sudiksha@localhost ~]$ bash reverse.sh
bash: reverse.sh: No such file or directory
[sudiksha@localhost ~]$ bash reverse.sh
Enter a number
123
Reversed number: 321
[sudiksha@localhost ~]$
```

```
#!/bin/bash
echo " Enter limit "
read limit
a=0
b=1
echo " Fibonacci Series "
echo " $a "
echo " $b "
for((i=3;i<=limit;i++))
do
    c=$((a+b))
    echo "$c"
    a=$b
    b=$c
done
```

```
[sudiksha@localhost ~]$ vi fibonacci.sh
[sudiksha@localhost ~]$ bash fibonacci.sh
Enter limit
21
Fibonacci Series
0
1
1235813213455891442333776109871597258441816765[sudiksha@localhost ~]$ vi fibonacci.sh
[sudiksha@localhost ~]$ bash fibonacci.sh
Enter limit
7
Fibonacci Series
0
1
1
2
3
5
8
[sudiksha@localhost ~]$
```